

6 SUPERSTRUCTURE COMPONENTS

6.1 GENERAL

This Chapter contains information and criteria related to the design, reinforcing, detailing, and construction of bridge superstructure elements and includes deviations from **LRFD**. This chapter covers erection schemes, beam and girder stability requirements, railings, curbs, joints, bearings, and deck drains. For additional information on concrete beams, decks, and steel girders, see **SDG** Chapter 4 and **SDG** Chapter 5.

6.2 CURBS AND MEDIANS (LRFD 13.11)

- A. For bridge projects that utilize curbs, match the curb height and batter on the roadway approaches.
- B. When the roadway approaches have a raised median, design the bridge median to match that on the roadway.

6.3 TEMPERATURE MOVEMENT (LRFD 3.12.2)

For all bridges other than longitudinally post-tensioned, segmental concrete bridges, calculate movement due to temperature variation (range) with an assumed mean temperature of 70 degrees Fahrenheit at the time of construction. Base joint and bearing design on the expansion and contraction for temperature ranges of **SDG** Table 2.7.1-1.

6.4 EXPANSION JOINTS

- A. For new construction, use only expansion joint types listed in Table 6.4-1.
- B. When an expansion joint is required, use one of the standardized expansion joints or details if possible. When a non-standardized expansion joint is required (e.g. finger joints and modular joints), design the joint using the following criteria:

Table 6.4-1 Expansion Joint Width Limitations by Joint Type

Expansion Joint Type	Maximum Open Width "W" (measured in the direction of travel at deck surface)
Hot Poured or Poured Joint without Backer Rod	3/4-inch
Poured Joint with Backer Rod	3-inches
Armored Strip Seal (Single gap)	Per LRFD 14.5.3.2
Modular Joint (Multiple modular gaps)	Per LRFD 14.5.3.2
Finger Joint	Per LRFD 14.5.3.2

Commentary: The joint types listed in Table 6.4-1 have historically been the most reliable and cost effective to construct and maintain. The maximum open width for the poured joint with backer rod is limited to 3-inches to reduce the potential for the joint seal material to be damaged by debris being pushed down into the joint from tires.